

EPPS 6354 Information Management

Assignment 4

Answer 1, 2 and two in 3 (i.e. 3a, 3b | 3b, 3c | 3a, 3c).

Prepare your answers in presentation format and be ready to present in class

1. Explain the difference between a weak and a strong entity set. Use an example other than the one in Chapter 6 to illustrate. (Consult Ch. 6, 6.5.3)
2. Design an E-R diagram for keeping track of the scoring statistics of your favorite sports team. You should store the matches played, the scores in each match, the players in each match, and individual player scoring statistics for each match. Summary statistics should be modeled as derived attributes with an explanation as to how they are computed. (Consult: [Practice Exercise solutions on textbook website](#))
 - a) Draw the E-R diagram using draw.io. Read this [website](#) for instructions.
 - b) Expand to all teams in the league (Hint: add team entity)

3. SQL exercise:

- a) Consider the query

```
select course_id, semester, year, sec_id, avg (tot_cred)
from takes natural join student
where year = 2017
group by course_id, semester, year, sec_id
having count (ID) >= 2;
```

- i. Explain why appending **natural join** *section* in the **from** clause would not change the result. (Consult Ch. 4, 4.1.1)
- ii. Test the results using the Online SQL interpreter (<https://www.db-book.com/university-lab-dir/sqljs.html>)
- b) Write an SQL query using the university schema to find the ID of each student who has never taken a course at the university. Do this using no subqueries and no set operations (use an outer join). (Consult Ch. 4, 4.1.3)
- c) Consider the following database, write a query to find the ID of each employee with no manager. Note that an employee may simply have no manager listed or may have a *null* manager (use natural left outer join). (Consult Ch. 4, 4.1.3)

```
employee (ID, person_name, street, city)
works (ID, company_name, salary)
company (company_name, city)
manages (ID, manager_id)
```
